

Abhigyan Majumdar

PHD IN SYSTEM DYNAMICS AND CONTROLS - UC DAVIS

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Skills

Engineering

Neural Networks, Model Predictive Control (MPC), Hardware-in-Loop (HIL) Testing, Reduced Order Modeling, System Identification, Koopman Operators

Software

Simulink & Simscape, Simulink RealTime, Vector CANalyzer, ANSYS (Mechanical, Fluent, Twin Builder), Arbin MITS Pro

General Programming

MATLAB, Python (NumPy, SciPy, Pandas, PyTorch, GPy, Matplotlib, CANlib), R, C++

Experience

UC Davis CORE Lab

Davis, California

GRADUATE STUDENT RESEARCHER

Sept 2024 - Present

- Working on multiple research projects, including - novel algorithms for data-driven Koopman Operator identification for general autonomous nonlinear dynamic systems; Predictive Safety Filtering for thermal derating of EV drive-units to enhance safety and drivability.

UC Davis EcoCAR Team

Davis, California

PROPULSION CONTROLS & MODELING LEAD

Sept 2022 - Aug 2024

- Built and led a multi-disciplinary team of around 15 undergraduate students to overhaul the powertrain of a Cadillac LYRIQ for the EcoCAR EV Challenge.
- Led powertrain architecture studies and propulsion supervisory controller development from scratch using Model-Based Systems Engineering workflows in MATLAB-Simulink powered MIL, HIL and VIL environments.
- Implemented functional safety by focusing on hazard analysis, requirements traceability, code coverage and multi-level (unit to VIL) testing for all features.

Farasis Energy Inc.

San Francisco Bay Area

BATTERY MODELING INTERN

June 2022 - Sept 2022

- Worked on reduced-order modeling and coupled electro-thermal simulation of Lithium-ion battery packs using ANSYS Fluent and TwinBuilder. Established the workflow for system-level simulations that can evaluate different fast-charging and thermal control algorithms based on battery performance.
- Work was presented at the SAW WCX 2023 - <https://doi.org/10.4271/2023-01-0931>

Green Technology Laboratory, UC Davis

Davis, California

RESEARCH TRAINEE

Feb 2022 - June 2022

- Assisted other graduate students in cycling, testing and characterization of Lithium-ion cells on Arbin battery testing stations. Processed experimental data using MATLAB for model parameter estimation.

Bio-Heat Transfer Laboratory

Rourkela, Odisha

UNDERGRADUATE RESEARCHER

Aug 2020 - May 2021

- Developed combined analytical-numerical solution method for phase change heat transfer in living tissue during cryosurgery, under the guidance of Prof. Sumit Kumar, NIT Rourkela. Reduced solution while preserving accuracy. Paper presented in 5-6th Thermal and Fluids Engineering Conference (2021). <https://doi.org/10.1615/TFEC2021.bio.036282>.

Jadavpur University

Kolkata, West Bengal

BATTERY MODELING INTERN

June 2019

- Carried out numerical simulation using Simulink to compare the thermal response of fresh and aged LFP cells by lumped modelling of electrochemical heat generation reactions. Compared PCM and heat pipe based passive cooling technologies to judge their effectiveness in delaying the onset and severity of thermal runaway.

Projects

Sliding Mode Wheel Slip Controller for AWD Electric Vehicle

UC DAVIS

Jan - Oct 2023

- Developed a non-linear Sliding Mode Controller with unique control law for a modified all-wheel-drive Cadillac LYRIQ. Reduced emergency braking distance and improved vehicle stability in low-mu conditions.
- Validated functionality in presence of noisy feedback in a MIL environment and published paper in the ASME Letters in Dynamic Systems and Controls - <https://doi.org/10.1115/1.4064346>

Energy Optimal Dual Stage Adaptive Cruise Controller for EVs

UC DAVIS

Apr - Jun 2023

- Designed a Non-linear Model Predictive Controller (NMPC) to optimize energy efficiency and passenger comfort in a longitudinal adaptive cruise controller with varying traffic and lead car dynamics.
- Validated real-time feasibility with MATLAB's fmincon solver, ensuring compliance with energy, comfort, and safety constraints in a model-in-loop (MIL) environment.

Autonomous Aerial Landing of Quadcopter

UC DAVIS

Apr - Jun 2022

- Built a proof-of-concept simulation environment in MATLAB-Simulink for autonomous landing of a UAV on a flying target. Developed an analytical trajectory generation algorithm for producing dynamically feasible, fast target following and landing. Flight stability and reference tracking were achieved through an adaptive LQR controller to handle the non-linear system dynamics.

Robust MIMO Controller for EV Cooling System

UC DAVIS

Apr - Jun 2022

- Developed a MIMO controller for an integrated EV cabin, battery, and traction motor cooling system, using the Youla Paraterimization, and H-Infinity controller design techniques respectively. Control design and simulation were done using MATLAB-Simulink and the controllers displayed robust disturbance rejection capability.

Education

University of California Davis

Davis, California

PHD IN MECHANICAL AND AEROSPACE ENGINEERING

September 2021 - present

- GPA: 3.92/4.0

National Institute of Technology (NIT) Rourkela

Rourkela, Odisha

BACHELOR OF TECHNOLOGY IN MECHANICAL ENGINEERING

July 2017 - May 2021

- GPA: 8.97/10

South Point High School

Kolkata, West Bengal

ALL INDIA SENIOR SCHOOL CERTIFICATE EXAMINATION (12TH GRADE)

April 2015 - May 2017

- Percentage: 90.2

Awards

First Place Propulsion Controls & Modeling Presentation Award

ECO CAR EV CHALLENGE YEAR 2

May 2024

- Presented to industry professionals from General Motors, MathWorks, US Department of Energy, Bosch, dSPACE and more regarding development and validation of EV propulsion controls software. Ranked first place amongst 15 universities across North America.

Over the HIL Award

ECO CAR EV CHALLENGE YEAR 2

May 2024

- Ranked first place among 15 universities from across North America for the best Hardware-in-the-Loop setup built using Speedgoat hardware for testing and validation of EV Propulsion Supervisory Controller code.